VIRTUAL REALITY: JUST WHAT THE DOCTOR ORDERED

By Zain Hamdia



You're standing on a bridge enjoying the view of a waterfall when suddenly you hear a noise behind you. You turn to see a man swinging a sword in your direction. You jump out of the way just in time and start to run. He chases you through a forest. Eventually he corners you inside of a cave. As you prepare to die, you hear a voice.

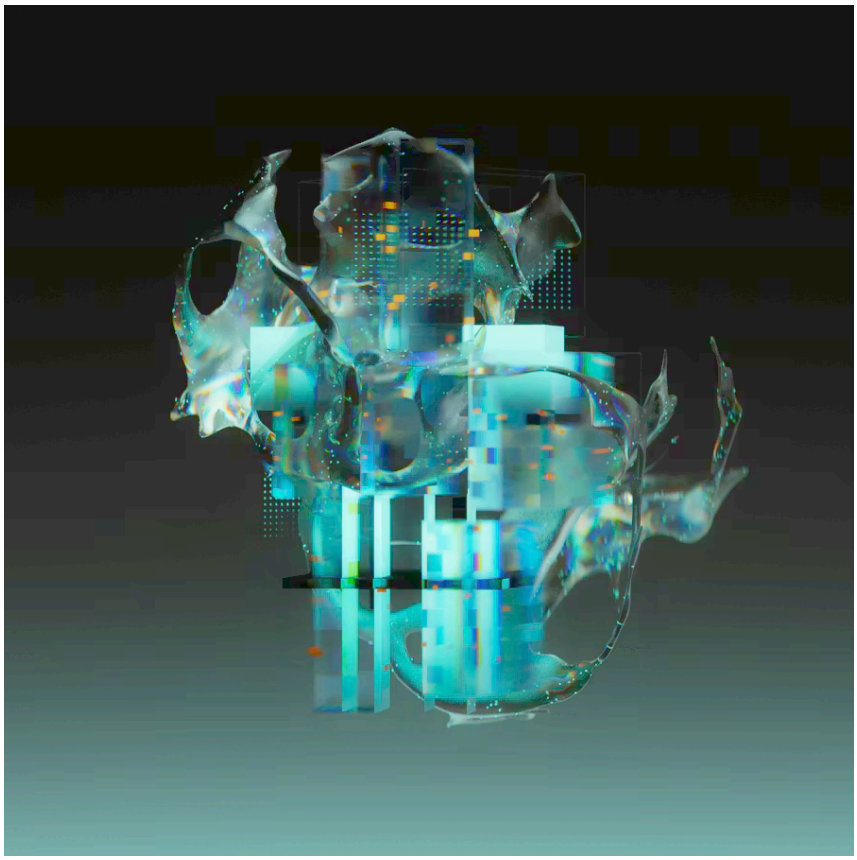
"Your time is up."

You slowly remove the virtual reality goggles, abruptly back to your hospital room. The doctor has an inquisitive look on his face.

"Do you feel any better?"

"Yes, I do," you reply.

From Recreation to Revolution



Virtual Reality (VR) has existed in some form beginning in the 1950s. Since that time and for most of its existence, VR has been used mainly for recreational purposes. But more recently, it has been at the forefront of something much more surprising and significant – the medical industry.

Cobbling through the decades with clunky devices and archaic simulators in theater-like settings, advancement in the technology has progressed quickly in the last 10 years.

One of the first VR headsets was made by Ivan Sutherland, a Harvard professor, in 1968. It was called the “Sword of Damocles” because the headset was so heavy, it had to be hung from the ceiling. Sutherland wanted people to feel like they were in a movie. It simulated driving a motorcycle through a city. The user could feel the vibrations that would come from a motorcycle, creating a much more immersive experience.

A New Tool for Pain Management

Today, virtual reality may literally be what the doctor ordered for pain management. One of its initial applications for this purpose was in the mid 90s. A psychologist named Hunter Hoffman took two teens that were undergoing burn wound treatment and put them in a standard VR shooting game while changing their wound dressings. With the help of VR, pain and anxiety levels decreased. He found that after repeating this experiment several times, the usage of VR had reduced most of the pain while patients remained unmedicated.

Approximately 50 million American adults live with some form of chronic pain. Virtual reality may be the solution. The U.S. Food and Drug Administration (FDA) recently approved the marketing of a virtual reality system called EaseVRx. The FDA evaluated a study using EaseVRx and the results were significant.

When the virtual reality treatment concluded, 66% of the users reported a 30% or more decrease in pain levels. After the treatment was completed, the subjects were checked on one month later and they all still reported a 30% pain decrease. After 3 months of checking with the subjects, the amount of pain reduction had stayed the same. With results like this, VR treatment could vastly improve the quality of someone's life.

Many hospitals such as Cedars-Sinai Medical Center, New York-Presbyterian Hospital and Boston Children's Hospital are using the VR treatment with high success rates.

How Does it Work?

But how does it work? There does not yet seem to be a clear answer, but there are two schools of thought.

One explanation is based on distraction of the mind. Most VR games being utilized for this purpose are skill-based games that engage the user's senses and focus, while also including different types of sensory feedback such as vibrations and sounds. These games can distract the brain and relieve pain without the need for medication.

The other explanation is that virtual reality relieves pain by changing the way the body processes a pain signal creating a state of analgesia, or blocking of pain, for the patient. Scientists theorize it is one, the other or some combination of the two.

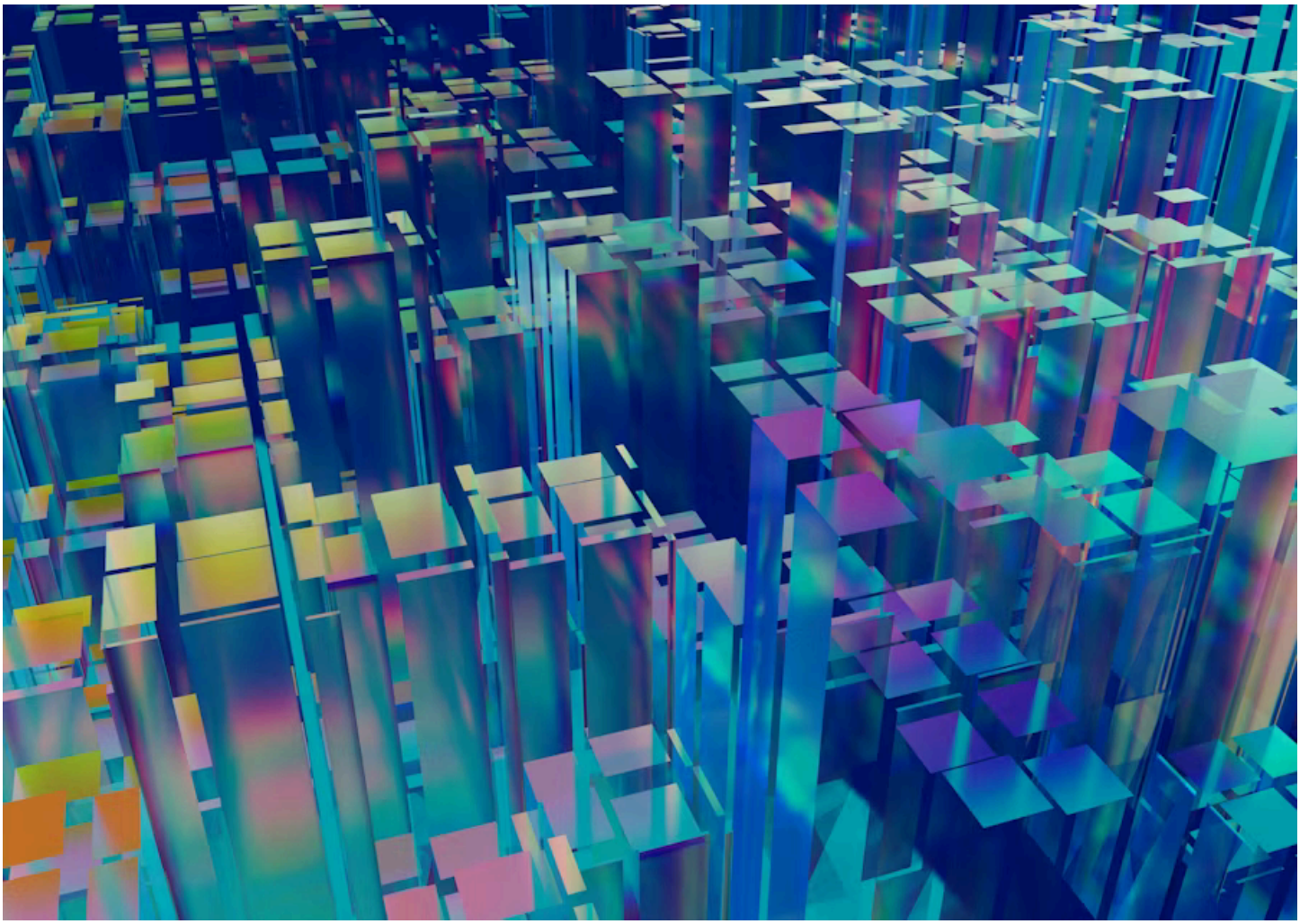
The Advantages of Virtual Reality Treatment

One of the many advantages of virtual reality is that it can treat different types of pain. Labor, chronic, acute, burn, and procedural pains to name just a few.

Another advantage is that it could improve mental health. Being in pain for a long period of time can take a toll on mental health. Ultimately managing physical health will benefit mental health and state of mind.

Virtual reality also does not require medication, resulting in the lack of chemical side effects.

One last, but crucial, advantage is that the structure of the headset allows for portability, so it can be taken anywhere. This is important in the aftermath of a pandemic, a time when hospitals were overcrowded with people infected with COVID 19. On a more positive note, the versatility of the headset allows it to be taken on vacations or simply used at home.



The One Major Drawback

Although it has a wide spectrum of pain it can help with, it does come with one main downside. When new users play VR for the first time, it is common that after about 10 minutes they start to feel nauseous, disoriented, get a headache and become dizzy. This is known as cyber-sickness and most users have experienced it at some point or another. Still vastly safer than chemical alternatives.

A Gamechanger in Healthcare

Virtual reality has had a long journey since its creation. What was once a machine used for entertainment purposes has evolved into an advanced, cutting-edge technology. Quite literally a game changer.

So next time you need to get a shot, make sure you ask for those virtual reality goggles!